IEOR 221 - HW 11 - Jie Wen

1

Using the Arbitrage Pricing Theory (APT), the required (equilibrium) expected return is

$$E(r) = r_f + \beta_I \lambda_I + \beta_R \lambda_R + \beta_C \lambda_C.$$

Given parameters:

- $r_f = 6\%$
- $\beta_I = 1.0,\ \lambda_I = 6\%$
- $\beta_R = 0.5,\ \lambda_R = 2\%$
- $\beta_C = 0.75,\ \lambda_C = 4\%$

Substituting these values:

$$E(r) = 6\% + 1.0 \cdot 6\% + 0.5 \cdot 2\% + 0.75 \cdot 4\% = 16\%$$

From the given return-generating process

$$r = 15\% + 1.0I + 0.5R + 0.75C + e,$$

and assuming the factors and e have zero mean, the actual expected return implied by the current price is

$$E(r) = 15\%.$$

The APT-required equilibrium return for this risk exposure is 16%. The stock offers only 15%, which is insufficient for its level of systematic risk $\Rightarrow$ it is **over-priced**

2

To achieve a market-neutral position, the overall beta of the portfolio must be zero.

The beta exposure of the futures position is given by:

$$\text{Total beta exposure} = N_A V_A \beta_A - N_B V_B \beta_B + N_I V_I \beta_I$$

Assuming the index future has $\beta_I \approx 1$, the equation is set to zero:

$$N_A V_A \beta_A - N_B V_B \beta_B + N_I V_I = 0$$

Solving for N_I:

$$N_I = \frac{N_B V_B \beta_B - N_A V_A \beta_A}{V_I} = \frac{110{,}000 - 80{,}000}{1250} = \frac{30{,}000}{1250} = 24$$